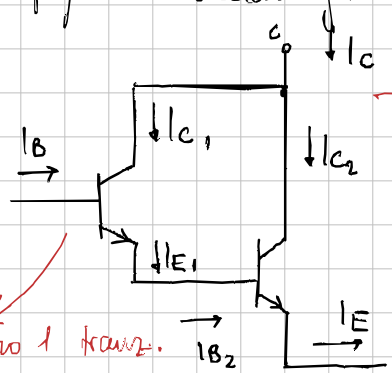
KASKADNA

Darlingtov spoj tranzistora

→ kod BipT najvažniji parametar je β; jako bitno da je što veći

↳ možemo pojačati dodavanjem tranzistora



→ možemo ih spojiti jer je visoko polarizirani pn-spoj

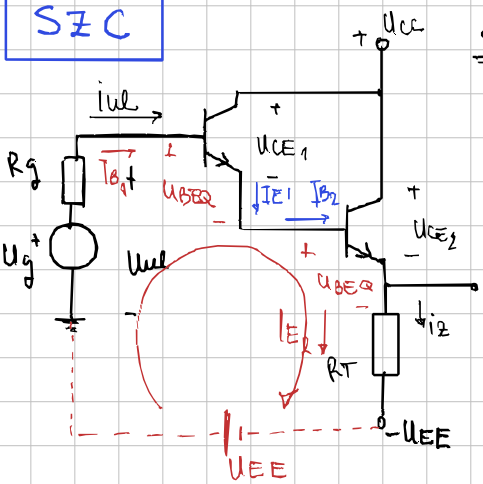
izvana vidimo samo 1 tranz.

$$I_{E_1} = \underset{(1)}{I_B} + I_{C_1} = I_B(1+\beta_1)$$

$$I_{C_2} = I_{B_2}\cdot\beta = I_{E_1}\cdot\beta = I_B(1+\beta_1)\beta_2$$

$$\approx > I_{C_2} = I_{B_1}\cdot\underbrace{\beta_1\cdot\beta_2}_{\text{novi } \beta \to \text{veći!}}$$

SZC



STATIKA

$$-U_{EE} + R_g\cdot I_{B_1} + U_{BEQ}\cdot 2 + I_{E_2}\cdot R_T = 0$$

$$I_{E2} = I_{B_2} + I_{C_2} = (\beta+1)\, I_{B_2}$$

$$I_{E_1} = I_{B_2}$$

$$I_{E_1} = I_{B_1} + I_{C_1}$$

$$I_{B_2} = (1+\beta)\, I_{B_1}$$

$$-U_{CC} + U_{CEQ_1} + U_{BEQ_2} + R_T\cdot I_{EQ_1} - U_{EE} = 0$$

$$-U_{CC} + U_{CEQ_2} + R_T\cdot I_{EQ_1} - U_{EE} = 0$$

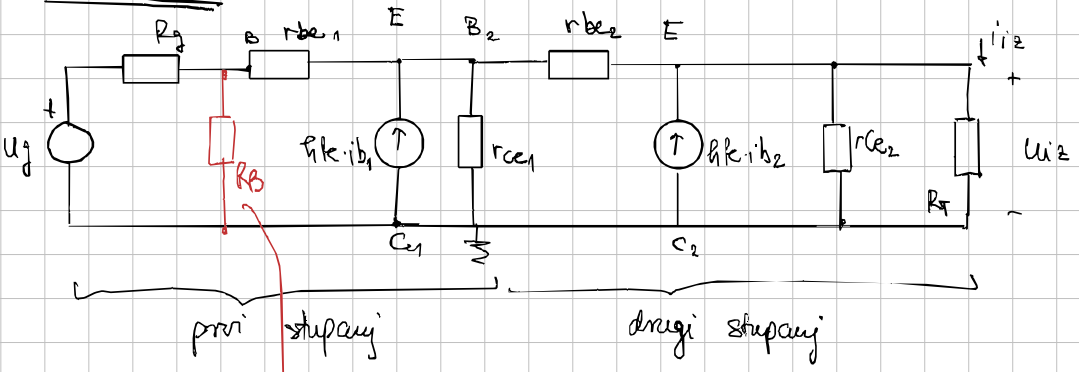
$U_{CEQ} > U_{BEQ}$ ... NAP

→ jako veliki Rul (~MΩ)

→ mali Riz (~10Ω)

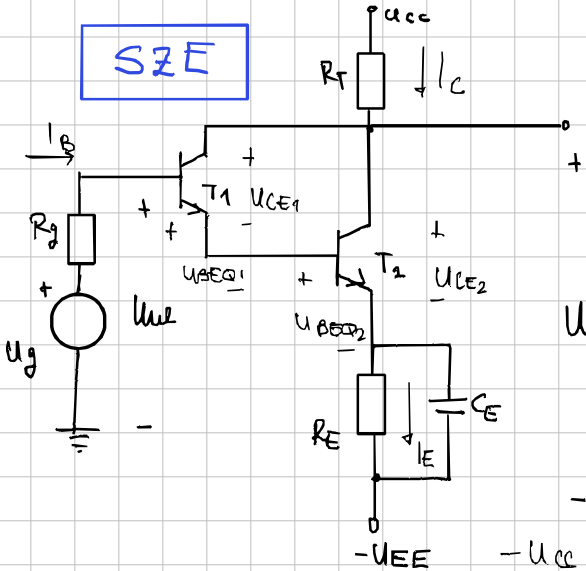
→ Av < 1, ≈ 1

DINAMIKA



gubimo snagu jako velikog Rul!

SZE



STATIKA

$$I_B \cdot R_g + U_{BEQ} \cdot 2 + I_E \cdot R_E - U_{EE} = 0$$

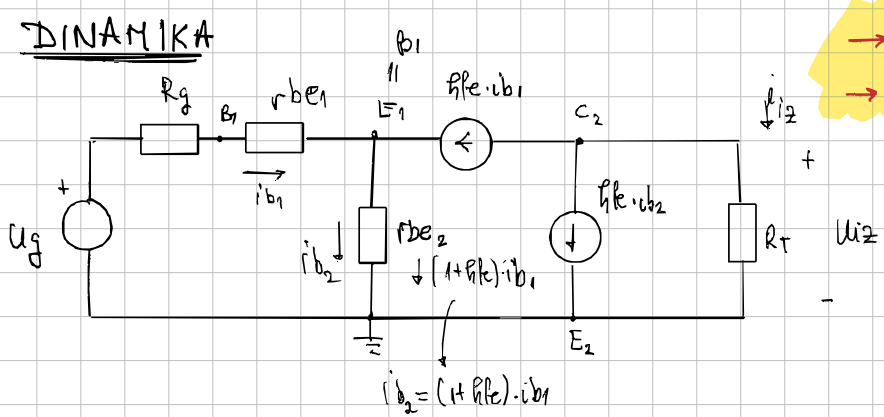
$$I_E = I_{E1} \cdot (\beta + 1) = (I_{B1} + I_{C1})(\beta + 1)$$

$$I_E = I_{B(1)}(1+\beta)(1+\beta)$$

$$-U_{CC} + R_T \cdot I_{CC} + U_{CEQ2} + I_E \cdot R_E = 0$$

$$-U_{CC} + R_T \cdot I_{CC} + U_{CEQ1} + U_{BEQ2} + R_E \cdot I_E = 0$$

DINAMIKA



→ R_{ul} ($\approx 50k\Omega$)

→ A_V ($\approx 100-200$)

→ SZC ima znatno veći R_{ul} i A_I

→ SZC A_V je dvostruko manji od običnog SZE

Općenito

1. stupanj = SZC sa pozitivnim strujnim i naponskim pojačanj. >0

2. stupanj = SZE pojačanja <0

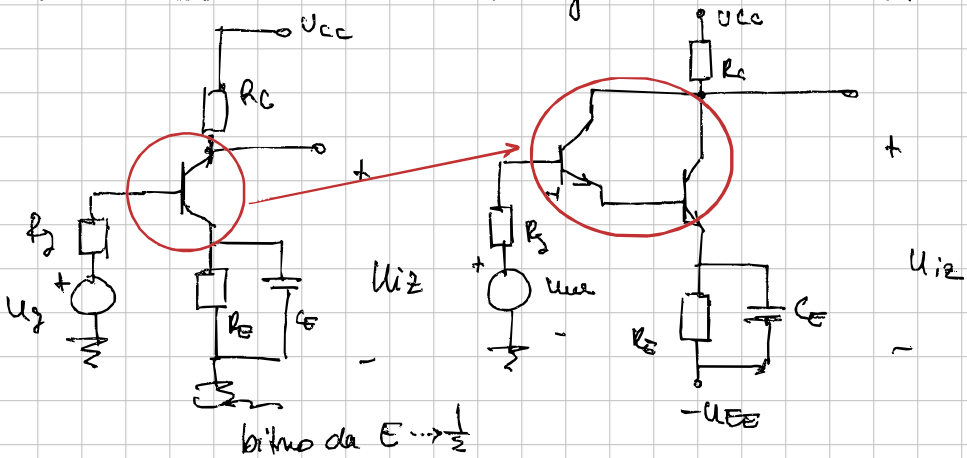
SULIGOJ 2025

1. Darlington → skica

→ SZE → umjesto T ide kaskadni

→ čemu služi

SZE inače: → zamjeni T sa kaskadnim



→ Želimo veći faktor β; jako getki veliki β pa kaskadnim spajanjem dobijemo za posljedicu da novonastali spoj radi kao tranzistor s $\beta = \beta_1 \cdot \beta_2 \cdots \beta_n$